

Methods of Data Modeling

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Simple regression

regression equation,

$$y = a + bx \quad \text{--- (1)}$$

y = dependent variable

x = independent variable

a = y -intercept

b = slope of regression line or
regression coefficient of y on x

Calculation of 'a' and 'b'

By solving two normal equations

$$\sum y = na + b\sum x \quad \text{--- (I)}$$

$$\sum xy = a\sum x + b\sum x^2 \quad \text{--- (II)}$$

Note: $y = a + bx$

i) Direct take summation

$$\sum y = \sum a + \sum bx$$

$$\therefore \sum y = na + b\sum x$$

ii) Multiply both sides by x and take $\sum$

$$xy = ax + bx^2$$

$$\sum xy = \sum ax + \sum bx^2$$

$$\therefore \sum xy = a\sum x + b\sum x^2$$

Steps for calculations:

i) r (correlation coefficient)

$$r = \frac{n \sum XY - \sum X \cdot \sum Y}{\sqrt{n \sum X^2 - (\sum X)^2} \sqrt{n \sum Y^2 - (\sum Y)^2}}$$

ii) Regression eqⁿ of Y on X ,

$$Y = a + bX \quad \text{--- (1)}$$

calculation of a and b

$$\sum Y = na + b \sum X$$

$$\sum XY = a \sum X + b \sum X^2$$

Standard Error of Estimate

$$SE = \sqrt{\frac{\sum Y^2 - a \sum Y - b \sum XY}{n-2}}$$

Test of regression coefficient

To test $H_0: b = 0$ (significant regression coefficient is not significant)

$H_1: b \neq 0$ (significant)

The test statistic is,

$$t = \frac{\hat{b}}{\hat{S}_b} \sim t_{(n-k-1)}$$

where $\hat{b}$ = calculated regression coefficient

$$\hat{S}_b = \sqrt{\frac{S_E}{\sum (X - \bar{X})^2}}$$

$$S_E = \sqrt{\frac{\sum y^2 - a\sum y - b\sum xy}{n-2}}$$

Conclusion:

Cal. $|t| \leq$ Tab. $|t|$, H_0 accepted

cal. $|t| >$ Tab. $|t|$, H_1 accepted

$H_0: b = 0$ (Regression coefficient not significant)

$H_1: b \neq 0$ (" " significant)

Partial Correlation

Let X_1, X_2 and X_3 are three variables,

$r_{1.23}$ = partial correlation coefficient betⁿ X_1 and X_2 taking X_3 as constant

$r_{13.2}$ = partial correlation coefficient betⁿ X_1 and X_3 taking X_2 as constant

$r_{23.1}$ = partial correlation coefficient betⁿ X_2 and X_3 taking X_1 as constant.

Then,

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{1 - r_{13}^2} \sqrt{1 - r_{23}^2}}$$

$$r_{13.2} = \frac{r_{13} - r_{12} r_{23}}{\sqrt{1 - r_{12}^2} \sqrt{1 - r_{23}^2}}$$

$$r_{23.1} = \frac{r_{23} - r_{12} r_{13}}{\sqrt{1 - r_{12}^2} \sqrt{1 - r_{13}^2}}$$

• Coefficient of Partial Determination

$$\text{If } r_{12.3} = 0.8 \text{ then } r_{12.3}^2 = (0.8)^2 \\ = 0.64 (64\%)$$

This implies that 64% change in X_1 is due to X_2 and remaining 36% change in X_1 is due to other unseen factor.

• Multiple correlation

$R_{1.23}$ = multiple correlation coefficient betⁿ X_1, X_2 and X_3 taking X_1 as a dependent variable

$R_{2.13}$ = Multiple correlation coefficient betⁿ X_1, X_2 and X_3 taking X_2 as a dependent variable

$R_{3.12}$ = Multiple correlation coefficient betⁿ X_1, X_2 and X_3 taking X_3 as a dependent variable.

$$\text{Then, } R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12} r_{23} r_{13}}{1 - r_{23}^2}}$$

$$R_{2.13} = \sqrt{\frac{r_{12}^2 + r_{23}^2 - 2r_{12} r_{23} r_{13}}{1 - r_{13}^2}}$$

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$$R_{3.12} = \sqrt{\frac{r_{13}^2 + r_{23}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{12}^2}}$$

- Coefficient of Multiple determination

Eg: If $R_{1.23} = 0.9$ then $R_{1.23}^2 = (0.9)^2$
 $= 0.81$ (81%)

This implies that 81% change in X_1 is due to X_2 and X_3 and remaining 19% change in X_1 is due to other unseen factors.

Multiple regression

It is given by:

$$Y = a_1 + b_1X_1 + b_2X_2$$

The values of constant a_1 , b_1 and b_2 are obtained by solving following three normal equations:

$$\sum Y = na_1 + b_1\sum X_1 + b_2\sum X_2$$

$$\sum YX_1 = a_1\sum X_1 + b_1\sum X_1^2 + b_2\sum X_1X_2$$

$$\sum YX_2 = a_1\sum X_2 + b_1\sum X_1X_2 + b_2\sum X_2^2$$

- Coefficient of Multiple determination from regression

$$R_{y.12}^2 = \frac{a_1\sum Y + b_1\sum YX_1 + b_2\sum YX_2 - n(\bar{y})^2}{\sum Y^2 - n(\bar{y})^2}$$

where,

$$\bar{y} = \frac{\sum y}{n}$$

- Standard error of estimate for multiple regression:

$$\sigma_{y,2} = \sqrt{\frac{\sum y^2 - a_1 \sum y - b_1 \sum y x_1 - b_2 \sum y x_2}{n-3}}$$

- Total sum of squares, sum of squares for regression and sum of squares due to error:

$$\therefore \text{Total variation} = \text{Regression} + \text{Error}$$
$$SST = SSR + SSE$$

$$SST = \sum (y - \bar{y})^2$$

$$SST = SSR + SSE$$

$$SSR = \sum (\hat{y} - \bar{y})^2$$

$$SSE = \sum (y - \hat{y})^2$$